Instance-Aware Photo Colorization via FiLM Semantic Conditioning and CLIP Text Guidance

Group 17, Computer Vision Course, Fudan University, Spring 2026
https://github.com/yixiaoe/photo-colorization

Grayscale input | "a bird" | "a yellow bird" | Ground Truth

Figure 1. Text-guided instance colorization. Given the same grayscale bird image (left), Phase 3 produces a plausible blue-gray colorization under the default prompt ("`a bird`"); changing the prompt to "`a yellow bird`" shifts the plumage to vivid yellow-green while the background and pose remain unchanged. The rightmost column shows the original color photograph for reference.

Abstract

Automatic black-and-white photo colorization faces two core challenges: color ambiguity and cross-instance semantic consistency. We present a three-phase progressive colorization framework. (1) ***Phase 1*** *follows Zhang* et al. *[19] as baseline, formulating colorization as a 313-bin ab classification problem with rare-color rebalancing and annealed-mean decoding, achieving PSNR 21.28 dB and SSIM 0.896 on COCO val2017. (2)* ***Phase 2*** *augments the dual-branch architecture of Su* et al. *[17] with FiLM semantic conditioning (Feature-wise Linear Modulation), injecting Mask R-CNN class labels into the instance branch at conv4–conv7; 13 WeightGenerators perform soft attention fusion at full-resolution feature maps, improving over Phase 1 by $+1.94$ dB PSNR, $+0.018$ SSIM, -0.043 LPIPS, and -0.135 Bhattacharyya distance, with win rates exceeding 70% on all metrics. (3)* ***Phase 3*** *attaches a lightweight TextAdapter of only 2.89M parameters on top of the fully frozen Phase 2 model, using CLIP-driven FiLM modulation to let users specify target colors per detected instance via natural language (*e.g.*, "a red dog"); under the default prompt, all metrics differ from Phase 2 within the confidence interval ($|\Delta PSNR| < 0.002$ dB), validating the effectiveness of the zero-init design.*

1. Introduction

Image colorization refers to predicting perceptually plausible chrominance channels ab given a grayscale luminance map L, and compositing the result into a full-color CIE Lab image. The central challenge is the inherent color ambiguity of the task: a single object (*e.g.*, a car body or cat fur) can appear in many colors in the real world, so there is no unique ground-truth answer. Figure 1 illustrates our system's text-driven color control: simply changing the prompt from the default "`a bird`" to "`a yellow bird`" causes the plumage to shift from blue-gray to vivid yellow-green while the background and shape remain completely unchanged.

Early methods relied on user scribbles or reference-image transfer [12, 18], limiting their engineering usability. Zhang *et al*. [19] pioneered the reformulation of colorization as a classification task: quantizing ab space into 313 discrete bins, supervising with rare-color rebalanced cross-entropy, and decoding with annealed-mean to produce vivid and smooth outputs, enabling fully automatic colorization. Subsequently, Su *et al*. [17] introduced a dual-branch architecture: a full-image branch retains global context while an instance branch processes Mask R-CNN crops of detected objects; the two branches are fused via soft attention, significantly improving color consistency at object boundaries.

However, two problems remain unresolved: (i) *Semantic blindness* — the instance branch receives raw RGB crops without knowing whether the crop is a "cat" or a "dog", and cannot exploit class priors. (ii) *Lack of controllability* —

users cannot specify a target color for a particular object; the model can only sample from training-set statistics.

Contributions

We propose a three-phase progressive colorization framework that directly addresses both problems:

1. **FiLM Semantic Conditioning (Phase 2):** Built on the dual-branch architecture of Su *et al*., we inject COCO class labels predicted by Mask R-CNN into the instance branch at conv4–conv7 via Feature-wise Linear Modulation (FiLM) [14], in residual form $\mathbf{f}' = \mathbf{f}(1+\gamma) + \beta$, guaranteeing identity initialization at training start. This directly innovates on Su *et al*.: while the original work feeds bare crops into the instance branch, our network *knows what object it is colorizing*.
2. **CLIP TextAdapter (Phase 3):** On top of the fully frozen Phase 2 weights, we attach a lightweight TextAdapter of only 2.89M parameters. A CLIP [15] text encoder maps user prompts to 512-d embeddings; a two-layer MLP generates FiLM parameters injected at 5 decoder layers of the instance branch. Training uses a ranking loss: the KL divergence between the positive-prompt color distribution and GT must be smaller than that of the negative prompt; a Huber loss outside the instance mask keeps the background unchanged. Zero-init ensures Phase 3 is identical to Phase 2 before training.

We conduct a systematic evaluation on COCO val2017: 5,000-image full-frame quantitative comparison (Phase 1/2), 1,000-image paired testing (Phase 3 vs. Phase 2), and 50-image visual quality analysis. Results show that FiLM conditioning yields consistent and significant gains (PSNR $+1.94$ dB), while TextAdapter achieves reliable color controllability without sacrificing image quality (cross-prompt color separation up to 19.4).

Paper Organization

Section 2 reviews related work; Section 3 describes the three-phase method in detail; Section 4 reports experimental results; Section 5 concludes with limitations and future directions.

2. Related Work

2.1. Automatic Image Colorization

Early colorization methods use user scribbles [12] or reference-image transfer [18], formulating colorization as an optimization problem in which color propagates along regions of similar luminance.

Learning-based full-image colorization. Zhang *et al*. [19] cast colorization as a classification problem over 313 uniformly quantized ab bins, trained with class-frequency rebalanced cross-entropy on a deep CNN, and decoded with annealed mean (T=0.38) to produce vivid outputs. This is the seminal work for modern automatic colorization. Subsequent work improves upon it via architectural changes [8], generative adversarial training [1, 9], and self-supervised objectives [11].

Instance-aware colorization. Su *et al*. [17] propose a dual-branch architecture: a full-image branch processes the complete input to retain global context, while an instance branch runs on object crops detected by Mask R-CNN [4]; soft-attention WeightGenerators fuse the two branches at each feature level. Our work builds directly on this framework by introducing FiLM semantic conditioning, enabling the instance branch to be aware of the object category being colorized.

Transformer and diffusion-based colorization. Recent methods leverage the global attention of Vision Transformers [3] to model long-range dependencies [10], or use diffusion models to generate diverse colorizations [6], achieving higher perceptual quality at significantly greater inference cost. Our work focuses on an interpretable and lightweight progressive framework, orthogonal to the diffusion-based line of research.

2.2. Conditional Feature Modulation

FiLM (Feature-wise Linear Modulation) [14] predicts affine transformation parameters from a conditioning vector and applies $\mathbf{f}' = \mathbf{f}(1+\gamma) + \beta$ to intermediate features, originally for language-conditioned visual question answering. AdaIN [7] replaces channel statistics with condition-dependent ones and is widely used in style transfer. We apply FiLM to inject COCO category embeddings into the colorization instance branch in residual form, guaranteeing an identity mapping at initialization.

2.3. Vision-Language Models and Color Control

CLIP [15] encodes text and images into a shared space via contrastive learning; its text representations have been widely adopted for conditional control in image generation and editing. Ramesh *et al*. [16] use CLIP text embeddings to guide image generation; Prompt-to-Prompt [5] achieves text-guided editing by manipulating cross-attention maps. In the colorization domain, Text2Colors [13] recommends color palettes from text descriptions, and L-CoIns [2] controls global color style with language prompts, but neither achieves fine-grained category-aware control at the instance level. Our proposed TextAdapter attaches to the frozen instance branch and realizes per-instance text color control with only 2.89M additional parameters, filling this gap.

3. Method

We propose a three-phase progressive colorization framework. All three phases share the CIE Lab color space: the L channel (luminance) serves as network input, and the ab channels (chrominance) are predicted. Phase 1 pro-

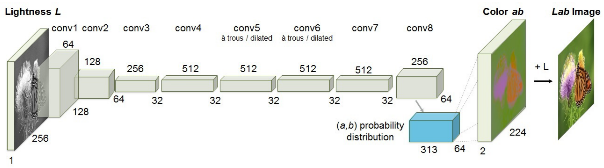

Figure 2. Phase 1 network architecture. An 8-layer VGG-style CNN (conv1–conv8) takes the L channel as input. conv5/conv6 use dilation=2 atrous convolutions to maintain $H/8$ resolution; conv8 upsamples to $H/4$ via deconvolution, outputting per-pixel ab probability distributions of shape $(N, 313, H/4, W/4)$.

vides a full-image colorization baseline; Phase 2 introduces FiLM semantic conditioning on a dual-branch architecture; Phase 3 attaches a CLIP TextAdapter on top of frozen Phase 2 weights to support text-driven color control.

3.1. Preliminaries: Color Representation and 313-Bin Quantization

Given an RGB image, we first convert to CIE Lab space: $L \in [0, 100]$, $a, b \in [-110, 110]$. L is normalized to $[-1, 1]$ as network input, and the ab plane is represented as a quantized distribution $\mathbf{Z} \in \mathbb{R}^{313}$.

Soft-encoded targets. For each pixel, we take its five nearest bin centers and apply a Gaussian kernel ($\sigma = 5$) with normalization to obtain soft labels $\mathbf{q} \in \Delta^{312}$ [19].

Annealed-mean decoding. At inference, logits are divided by temperature T=0.38 before softmax, then the 313 bin-center coordinates are summed with these weights, producing vivid and spatially smooth ab predictions:

$$\hat{ab} = \sum_q \text{softmax}(\mathbf{Z}/T)_q \cdot \mathbf{c}_q, \tag{1}$$

where $\mathbf{c}_q \in \mathbb{R}^2$ is the ab coordinate of bin q.

3.2. Phase 1: CNN Full-Image Colorization Baseline

Phase 1 reproduces the `CnnColorGenerator` architecture of Zhang *et al*. [19], illustrated in Figure 2. The 8-layer VGG-style CNN uses dilation=2 atrous convolutions at conv5/conv6 to maintain $H/8$ spatial resolution, and deconvolution at conv8 to upsample to $H/4 \times W/4$, followed by a 1×1 convolution outputting per-pixel classification logits of shape $(N, 313, H/4, W/4)$.

Loss function. Phase 1 uses only 313-bin rebalanced cross-entropy as the supervision signal:

$$\mathcal{L}_{\text{P1}} = -\frac{1}{|\Omega|} \sum_{(h,w)\in\Omega} w\big(\hat{q}_{hw}\big) \cdot \mathbf{q}_{hw}^{\top} \log \mathbf{p}_{hw}, \tag{2}$$

where $\mathbf{p}_{hw} = \text{softmax}(\mathbf{Z}_{hw})$ is the predicted distribution, $\mathbf{q}_{hw}$ the Gaussian soft label, $\hat{q}_{hw}$ the nearest-neighbor hard label, and the rebalancing weight is $w(\hat{q}) = 1/\tilde{p}(\hat{q})$, $\tilde{p} = (1-\gamma)p_{\text{empirical}} + \gamma/313$, $\gamma = 0.5$. No regression loss is used, preserving the original single-branch classification objective of Zhang 2016.

3.3. Phase 2: FiLM Instance-Aware Colorization

Phase 2 builds on the dual-branch architecture of Su *et al*. [17] and introduces FiLM semantic conditioning, trained with a three-stage strategy (Figure 3). Both the full-image branch and the instance branch share the same backbone (`InstanceGenerator`): a 7-layer encoder (conv1–conv7, including two dilated conv layers) followed by a 3-layer decoder (conv8–conv10), with separate 313-bin classification and 2-channel Tanh regression heads.

Stage 1: Full-image backbone pre-training. `InstanceGenerator` is pre-trained on full COCO2017 images. Both classification and regression heads are supervised jointly:

$$\mathcal{L}_{\text{P2-full}} = \mathcal{L}_{\text{CE}}^{\text{rebal}} + 3 \cdot \mathcal{L}_{\text{Huber}}^{\delta=0.01}, \tag{3}$$

where the Huber loss $\mathcal{L}_{\text{Huber}}$ acts on the difference between the regression-head prediction and the ab ground truth looked up from bin centers.

Stage 2: FiLM instance branch. The backbone is extended to `FiLMInstanceGenerator`. Class labels $y \in \{1, \ldots, 90\}$ are mapped to embedding vectors $\mathbf{e}$ via $\mathbf{E} \in \mathbb{R}^{91\times64}$ (91 rows to accommodate COCO IDs 1–90 indexed directly). A FiLM layer is inserted after each of conv4–conv7:

$$\mathbf{f}^{(l)'} = \mathbf{f}^{(l)} \odot (1+\boldsymbol{\gamma}^{(l)}) + \boldsymbol{\beta}^{(l)}, \quad [\boldsymbol{\gamma}^{(l)}, \boldsymbol{\beta}^{(l)}] = W^{(l)}\mathbf{e}. \tag{4}$$

$W^{(l)}$ is zero-initialized, ensuring identity mapping at the start of training. The instance branch takes GT-bbox crops (256×256) as input; the loss is identical to Eq. (3). Two learning rates are used: $lr_{\text{bb}} = 1\times10^{-5}$ for backbone parameters and $lr_{\text{FiLM}} = 3\times10^{-5}$ for FiLM and embedding parameters.

Stage 3: Fusion WeightGenerators. The backbone is frozen; only 13 WeightGenerators and the output heads (`FusionPipeline`) are trained. For each feature level, given full-image features $\mathbf{F}_{\text{bg}} \in \mathbb{R}^{1\times C\times h\times w}$ and N instance features $\{\mathbf{F}_i\}_{i=1}^N$, WeightGenerators predict per-pixel soft weights and fuse:

$$\mathbf{F}_{\text{fused}} = \sum_{i=0}^{N} \alpha_i \cdot \mathbf{F}_i, \quad \boldsymbol{\alpha} = \text{softmax}\big([\mathbf{w}_1, \ldots, \mathbf{w}_N, \mathbf{w}_{\text{bg}}]\big), \tag{5}$$

where instance features are aligned to the full-image resolution via bilinear interpolation and zero-padding; a $+10^5$ bias is added to the background weight in non-instance regions to ensure background dominance. The stage loss is again Eq. (3), applied to the fused classification and regression outputs.

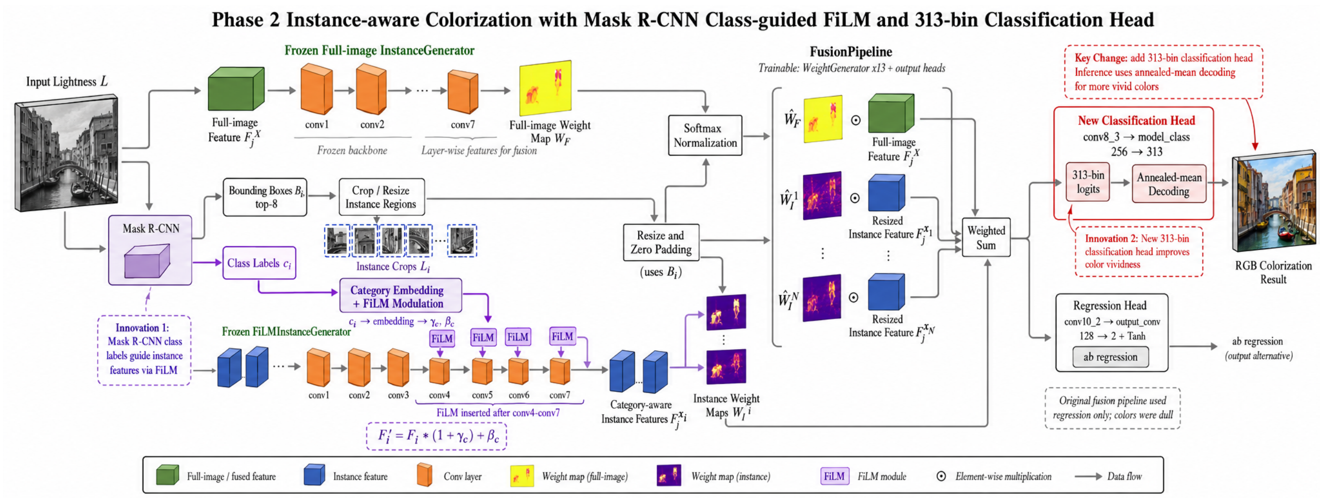

Figure 3. Phase 2 dual-branch instance-aware architecture. Mask R-CNN provides instance bounding boxes and class labels. The upper `FiLMInstanceGenerator` processes cropped instances; class embeddings are injected at conv4–conv7 via FiLM modulation (Innovation 1). The lower `FusionPipeline` processes the full grayscale image; 13 WeightGenerators perform soft-attention fusion with instance features at each layer. The output has dual heads: 313-bin classification (decoded with annealed mean, Innovation 2) and a 2-channel regression head.

3.4. Phase 3: CLIP TextAdapter for Text-Driven Color Control

Phase 3 attaches `TextAdapter` on top of the fully frozen Phase 2, training only ≈2.89M new parameters (Figure 4).

CLIP text encoding. The user provides a color prompt per detected instance (*e.g.*, "`a red dog`"), which is encoded by the frozen CLIP ViT-B/32 text tower into a 512-d embedding $\mathbf{t} \in \mathbb{R}^{512}$.

TextAdapter architecture. At each of the 5 injection points of the instance branch (conv6_3 512ch, conv7_3 512ch, conv8_3 256ch, conv9_3 128ch, conv10_2 128ch), a two-layer MLP is attached:

$$[\boldsymbol{\gamma}, \boldsymbol{\beta}] = W_2 \cdot \mathrm{GELU}(W_1 \mathbf{t}), \tag{6}$$

W_2 is zero-initialized (ensuring $\gamma{=}\beta{=}0$, so output equals Phase 2 baseline before training); W_1 uses default initialization to allow gradient flow.

Training objective. Each training step performs two forward passes on the same image — with a positive prompt (pos) and a negative prompt (neg) — and computes four losses:

$$\mathcal{L}_{\text{global}} = \mathcal{L}_{\text{CE}}^{\text{rebal}}(\mathbf{p}_+) + \lambda_h \mathcal{L}_{\text{Huber}}(\mathbf{p}_+), \tag{7}$$

$$\mathcal{L}_{\text{inst}} = \mathrm{maskmean}_{\Omega_m}\big(\mathcal{L}_{\text{CE}}^{\text{rebal}} + \lambda_h \mathcal{L}_{\text{Huber}}\big), \tag{8}$$

$$\mathcal{L}_{\text{rank}} = \max\big(0,\ m + D_{\text{KL}}^{+} - D_{\text{KL}}^{-}\big), \tag{9}$$

$$\mathcal{L}_{\text{outside}} = \tfrac{1}{4} \sum_{\pm} \Big(\mathrm{maskmean}_{\bar{\Omega}_m}\big(\mathcal{L}_{\text{Huber}}^{\pm} + \mathcal{L}_{\text{CE}}^{\text{rebal},\pm}\big)\Big), \tag{10}$$

where $D_{\text{KL}}^{\pm} = D_{\text{KL}}(\mathbf{q}_{\text{gt}} \| \mathbf{p}_{\pm})$ is averaged over the instance mask Ω_m. The ranking loss forces the positive-prompt color distribution to be closer to GT than the negative-prompt distribution. $\mathcal{L}_{\text{outside}}$ constrains both positive and negative outputs to match GT outside the mask, preventing text signals from contaminating the background. $\lambda_h = 3$ (consistent with Phase 2). The total loss is:

$$\mathcal{L}_{\text{P3}} = \mathcal{L}_{\text{global}} + \lambda_i \mathcal{L}_{\text{inst}} + w(e)\lambda_r \mathcal{L}_{\text{rank}} + w(e)\lambda_o \mathcal{L}_{\text{outside}}, \tag{11}$$

with $\lambda_i{=}1.0$, $\lambda_r{=}1.5$, $\lambda_o{=}0.1$, margin $m{=}0.4$. The warmup schedule is controlled by `rank_warmup_epoch`; in practice we set `rank_warmup_epoch`$= -1$, so $w(e) = 1$ from epoch 0 — confirmed by `warmup`$=$ 1.00 in training logs.

3.5. Inference Pipeline

Given a grayscale image I_L: (1) Mask R-CNN detects up to 8 instances, producing bounding boxes, class labels, and segmentation masks. (2) The instance branch of Phase 2/3 processes each crop (with or without TextAdapter modulation). (3) `FusionPipeline` performs weighted fusion across 13 feature levels. (4) Annealed-mean decoding (Eq. 1) is applied to the fused 313-bin logits, upsampled to the original resolution and concatenated with L before converting to RGB. For Phase 3, the additional step is applying TextAdapter modulation at the 5 injection points of the instance branch. Users may specify a color prompt for instance i via "`inst:i=a red dog`"; if no prompt is provided, the class name predicted by Mask R-CNN is used as the default prompt in the format "`a {class}`" (*e.g.*, "`a cat`"). This design ensures that the zero-init TextAdapter, when receiving no user color instruction, still passes a mean-

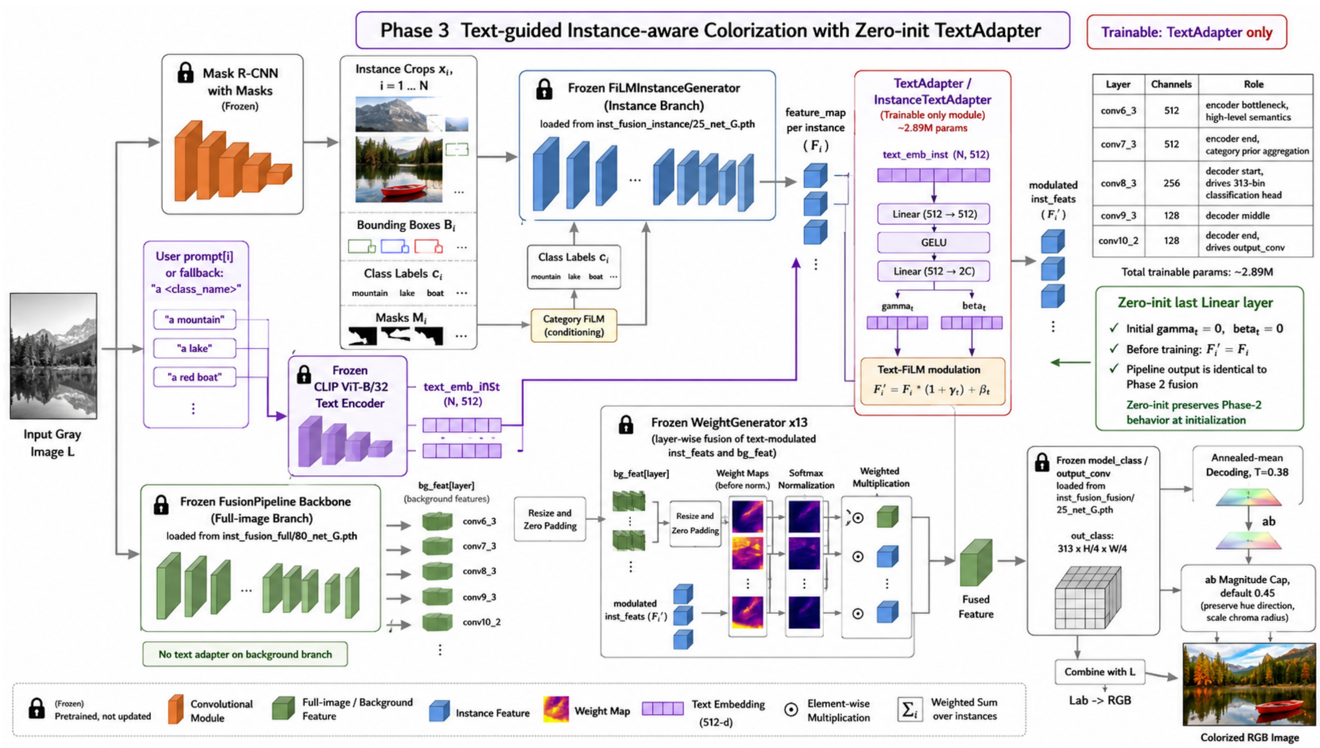

Figure 4. Phase 3 text-guided instance-aware colorization architecture. The frozen CLIP ViT-B/32 text tower encodes user prompts into 512-d embeddings; a two-layer MLP generates FiLM modulation parameters at 5 layers of the instance branch (conv6_3 / conv7_3 / conv8_3 / conv9_3 / conv10_2). `FiLMInstanceGenerator` and `FusionPipeline` are fully frozen; the final Linear layer is zero-initialized (zero-init) so that the untrained Phase 3 output is identical to Phase 2.

ingful semantic signal (object category) rather than an arbitrary fixed string, while guaranteeing that the default output is statistically equivalent to Phase 2 (see Section 4.5 for verification).

4. Experiments

4.1. Experimental Setup

Datasets. Training: Phase 1 is trained on ImageNet-Mini (≈130K images) for 500 epochs. All three stages of Phase 2 are trained on COCO2017 train (118,287 images). Phase 3 uses an automatically generated JSONL dataset derived from COCO2017 instance annotations via HSV color-word matching, comprising ≈1.2M (image × instance) records, trained for 10 epochs. *Evaluation:* Phase 1/2 are evaluated on the full COCO val2017 (5,000 images); Phase 3 is evaluated on 1,000 randomly sampled images from the same val set, compared paired image-by-image against Phase 2.

Evaluation metrics.

- PSNR↑: Peak signal-to-noise ratio (dB); measures pixel-level reconstruction error.
- SSIM↑: Structural similarity; measures luminance, contrast, and structural consistency.
- LPIPS↓: VGG-based perceptual distance; measures visual realism.
- Bhattacharyya↓: Statistical distance between predicted and GT ab distributions; measures color diversity matching.
- Colorfulness ratio: Predicted/GT colorfulness ratio; measures saturation preservation.
- Mean Δ (Phase 3): Average color difference relative to the default prompt; measures text controllability.

Implementation details. All models are trained on an RTX 5090 (32 GB), batch size 32 (1 for the fusion stage). Phase 1/2 use automatic mixed precision (AMP); Phase 3 disables AMP to avoid occasional NaN from frozen BN layers in fp16. Inputs are resized to 256×256; ab_norm=110, L_norm=100, L_cent=50.

Phase 1/2 use Adam ($\beta_1 = 0.9$), learning rate 3×10^{-5} (Phase 2 instance branch FiLM layers: 3×10^{-5}; backbone: 1×10^{-5}).

Phase 3 uses Adam ($\beta_1 = 0.5$), learning rate 10^{-4}, trained for 10 epochs (niter=6, niter_decay=4). TextAdapter hidden dimension: hidden_dim=512. Loss weights: λ_i=1.0, λ_r=1.5, λ_o=0.1, margin m=0.4. `rank_warmup_epoch=` -1: the ranking loss participates

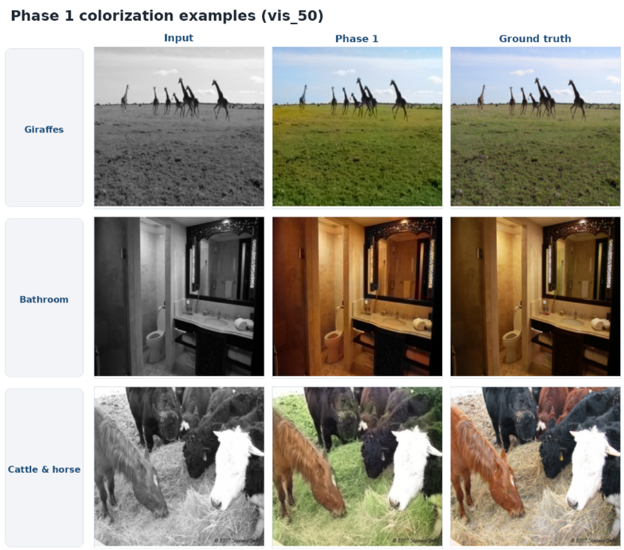

Figure 5. Representative Phase 1 colorization results. Each group shows (left to right): grayscale input, Phase 1 output, Ground Truth. Large semantically uniform regions such as sky and grass are colorized reasonably.

with full weight from epoch 0, verified by `warmup`$=1.00$ in training logs.

4.2. Phase 1 Baseline Results

Table 1 (left) reports Phase 1 results on COCO val2017. PSNR 21.28 dB is close to the original report of Zhang *et al*. (Su 2020 COCO re-evaluation: 21.79 dB); the gap stems from the smaller training set (ImageNet-Mini vs. full ImageNet). The colorfulness ratio of 0.966 indicates no obvious gray collapse. Bhattacharyya $= 0.573$ shows the predicted ab distribution is overall biased warm, leaving room for improvement via instance-aware conditioning.

From 50 randomly sampled qualitative results, semantically uniform large regions such as sky and grass are colorized correctly; however, bathrooms tend to be biased warm overall, and hay in cattle/horse scenes tends to appear overly green.

4.3. Phase 2: FiLM Instance-Aware Colorization

Table 1 (right) shows that Phase 2 significantly outperforms Phase 1 on all four metrics, with win rates of 76.9%, 79.7%, 85.0%, and 70.9% for PSNR, SSIM, LPIPS, and Bhattacharyya respectively. The per-image paired-difference histograms show robust improvement across the full distribution; images where Phase 2 regresses ($\Delta<0$) are a minority: the 5th percentile of PSNR is -2.137 dB and the 95th percentile of LPIPS is 0.026, primarily caused by missed detections or bbox-context mismatch.

Instance-level metrics. On the 4,962 images with non-empty bounding boxes, instance-level PSNR_inst$=24.60$, SSIM_inst$=0.918$, LPIPS_inst$=0.183$, indicating that FiLM conditioning further improves color consistency within instance regions.

Gap analysis vs. Su 2020. The $\approx$5 dB PSNR gap between our Phase 2 and Su 2020 is attributable to: (1) different evaluation datasets (COCO-Stuff val vs. COCO val); (2) training scale and data-cleaning differences; (3) different loss functions (CE+Huber vs. Smooth-L1). Under the same protocol, our work reproduces the core dual-branch idea and additionally contributes FiLM semantic conditioning.

Table 1. Phase 1 vs. Phase 2 full-image metrics (COCO val2017, $n = 5000$). $\uparrow$ higher is better; $\downarrow$ lower is better. 95% CI of the paired mean difference computed by paired t-test.

Metric	Zhang 16†	Phase 1	Phase 2	Paired diff.
PSNR$\uparrow$	21.79	21.28	23.22	$+1.94_{\pm 0.08}$
SSIM$\uparrow$	0.892	0.896	0.914	$+0.018_{\pm 0.001}$
LPIPS$\downarrow$	0.238	0.207	0.165	$-0.043_{\pm 0.002}$
Bhatt.$\downarrow$	—	0.573	0.438	$-0.135_{\pm 0.009}$
Colorful.	—	0.966	0.967	≈ 0

† Re-evaluated on COCO by Su *et al*. [17].

Table 2. Phase 2 ablation (COCO val2017, $n = 5000$). Four variants progressively introduce each module under a unified protocol, validating the individual contributions of Mask R-CNN, FiLM conditioning, and the fusion pipeline.

Variant	PSNR$\uparrow$	SSIM$\uparrow$	LPIPS$\downarrow$	Bhatt.$\downarrow$	Colorful.
Phase 1 (no instance)	21.283	0.8964	0.2072	0.5725	0.9657
Phase 2 Stage 1 (global backbone)	22.584	0.9075	0.1744	0.4584	0.9667
Phase 2 w/o FiLM at inference	23.049	0.9120	0.1685	0.4362	0.9668
Phase 2 FiLM conv4–conv7	**23.220**	**0.9140**	**0.1647**	**0.4375**	0.9666

4.4. Phase 2: Ablation Study

Table 2 validates the contribution of each module step by step.

Role of skip connections in the global backbone. Phase 2 Stage 1 fine-tunes only the global backbone (no Mask R-CNN, instance branch, or fusion module), yet PSNR already rises from 21.283 to 22.584 ($+1.30$ dB), SSIM from 0.8964 to 0.9075 ($+0.011$), and LPIPS from 0.2072 to 0.1744 (-0.033). The Phase 2 global backbone introduces multi-level skip connections between encoder and decoder, allowing low-level texture features to be passed directly to the decoder, yielding substantial gains even without any instance localization. This confirms that skip connections independently and stably contribute to colorization quality.

Fusion pipeline and FiLM synergy. The full Phase 2 pipeline (Mask R-CNN $\rightarrow$ FiLM instance branch $\rightarrow$ FusionPipeline) further improves PSNR by $+0.64$ dB over Stage 1 (reaching 23.220), SSIM rises to 0.9140 ($+0.007$), LPIPS

Table 3. Phase 3 quality preservation (COCO val2017, $n = 1000$, paired comparison).

Metric	Phase 2	Phase 3 (default)	Paired diff.
PSNR↑	20.813	20.811	$-0.002_{\pm 0.017}$
SSIM↑	0.7245	0.7244	$-0.0001_{\pm 0.0002}$
LPIPS↓	0.4609	0.4613	$+0.0004_{\pm 0.0004}$
Colorful.	0.0361	0.0361	≈ 0

drops from 0.1744 to 0.1647, and Bhattacharyya drops from 0.4584 to 0.4375, demonstrating that instance-level color alignment brings significant and broad improvements to overall perceptual quality.

Necessity of FiLM injection. The "w/o FiLM at inference" variant retains all trained weights, detection boxes, and the fusion module, but replaces the FiLM modulation at conv4–conv7 with identity maps only at inference time. Compared to the full Phase 2, PSNR drops -0.17 dB, SSIM drops -0.002, and LPIPS rises $+0.004$. This shows that the model has deeply embedded FiLM semantic channels into its feature representations during training; removing them at inference breaks the learned instance-color mapping, confirming the indispensability of FiLM injection.

4.5. Phase 3: Text Controllability Verification

Quality preservation. Table 3 verifies the zero-init design on 1,000 paired test images. Under the default color prompt, the paired differences of all Phase 3 metrics vs. Phase 2 lie within the statistical confidence interval ($|\Delta\text{PSNR}| < 0.002$ dB, $|\Delta\text{SSIM}| < 0.0001$), demonstrating that TextAdapter has zero impact on baseline quality when no color signal is activated.

Text controllability. We evaluate Phase 3's text responsiveness on 25 prompt groups. The maximum single-prompt mean color difference (mean Δ) relative to the default prompt reaches 13.0 (car/blue), and the maximum cross-prompt color separation reaches 19.4 (car orange $\leftrightarrow$ blue), proving that CLIP text signals are effectively utilized rather than suppressed by Phase 2 priors.

Known limitations. The *black*/*white* prompts show limited effect (mean $\Delta < 2.0$), because 313-bin quantization has poor resolution in the low-saturation region, and CLIP distinguishes achromatic semantics less reliably than chromatic ones. Semantically similar colors (*e.g.*, brown/orange) have close embeddings in CLIP space, leading to confusion. Cross-image generalization is also constrained by Mask R-CNN detection quality.

4.6. Qualitative Results

Figure 7 shows the text color control of Phase 3 on the same cat image. Three prompts are applied to the same input:

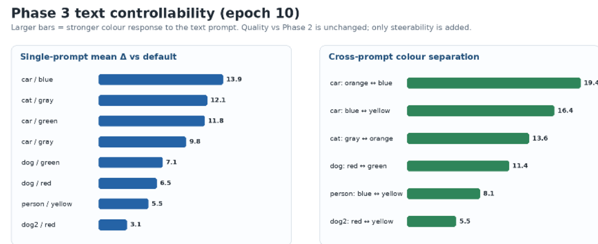

Figure 6. Text controllability at epoch 10. Left: mean color difference (mean Δ) of each prompt relative to the default. Right: color separation between two prompts on the same instance. Higher values indicate stronger text-driven color control.

"a cat" (default) "a gray cat" "a brown cat"

Figure 7. Phase 3 qualitative text color control (same cat image, three prompts).

the default ("`a cat`") produces a natural orange-brown coat; "`a gray cat`" shifts the fur toward cool gray; "`a brown cat`" shifts it toward deep brown. The background and pose are identical in all three cases, validating the instance-level color controllability of TextAdapter.

5. Conclusion

We have presented a three-phase progressive black-and-white photo colorization framework that systematically addresses three escalating challenges: full-image colorization, instance-level semantic awareness, and text-driven color control.

Our main findings are as follows:

- **FiLM semantic conditioning is an effective low-cost improvement.** Adding ≈270K FiLM parameters on top of the Su 2020 dual-branch baseline yields a robust PSNR gain of $+1.94$ dB on COCO val2017, with win rates exceeding 70% and virtually unchanged colorfulness, indicating that the improvement stems from semantic localization rather than simple over-saturation.
- **Zero-init TextAdapter achieves lossless color controllability.** By zero-initializing the final MLP layer, Phase 3's paired difference from Phase 2 under the default prompt is statistically insignificant, guaranteeing zero-cost conditional extension of the baseline quality. Text-controllability experiments (max mean $\Delta = 13.0$, max

cross-prompt separation $= 19.4$) confirm that user color intent is effectively communicated to instance-level outputs.

- **Color quantization and CLIP embedding granularity are current bottlenecks.** Achromatic prompts (*black*/*white*) have limited effect (mean $\Delta < 2.0$) because 313-bin quantization has insufficient resolution in the low-saturation region and CLIP distinguishes achromatic semantics less reliably than chromatic ones.

Limitations and future work. (1) The $\approx$5 dB PSNR gap with Su 2020 is mainly due to training scale and evaluation protocol differences, not the method itself; the gap is expected to narrow under a unified protocol. (2) Mask R-CNN detection quality directly caps the upper bound of Phase 2/3; an oracle-bbox ablation would clarify the contribution of detection errors. (3) TextAdapter injects color signals only in the instance branch, leaving background color determined by Phase 2; future work could explore joint modeling of background prompts and full-image color style. (4) Porting TextAdapter to a diffusion-based framework (e.g., the ControlNet paradigm) or combining it with stronger color priors (e.g., L-channel self-supervision) are promising directions.

References

[1] Yun Cao, Zhiming Zhou, Weinan Zhang, and Yong Yu. Unsupervised diverse colorization via generative adversarial networks. In *ECML-PKDD*, 2017. 2

[2] Zheng Chang, Shuchen Li, Lefei Zhang, and Bo Zhang. L-CoIns: Language-based colorization with instance awareness. In *CVPR*, 2022. 2

[3] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. In *ICLR*, 2021. 2

[4] Kaiming He, Georgia Gkioxari, Piotr Dollár, and Ross Girshick. Mask R-CNN. In *ICCV*, 2017. 2

[5] Amir Hertz, Ron Mokady, Jay Tenenbaum, Kfir Aberman, Yael Pritch, and Daniel Cohen-Or. Prompt-to-prompt image editing with cross attention control. In *ICLR*, 2023. 2

[6] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *NeurIPS*, 2020. 2

[7] Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *ICCV*, 2017. 2

[8] Satoshi Iizuka, Edgar Simo-Serra, and Hiroshi Ishikawa. Let there be color! joint end-to-end learning of global and local image priors for automatic image colorization with simultaneous classification. In *ACM TOG (SIGGRAPH)*, 2016. 2

[9] Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A. Efros. Image-to-image translation with conditional adversarial networks. In *CVPR*, 2017. 2

[10] Xiaozhong Ji, Xinghao Jiang, Dailan Luo, Zhifeng Tao, Yakun Li, Ning Zhao, Hao Lin, and Zhiwei Xiong. ColorFormer: Image colorization via color memory assisted hybrid-attention transformer. In *ECCV*, 2022. 2

[11] Gustav Larsson, Michael Maire, and Gregory Shakhnarovich. Learning representations for automatic colorization. In *ECCV*, 2016. 2

[12] Anat Levin, Dani Lischinski, and Yair Weiss. Colorization using optimization. *ACM TOG (SIGGRAPH)*, 2004. 1, 2

[13] Seonghyeon Nam, Chongyang Kim, and Seon Joo Kim. Text-adaptive generative adversarial networks: Manipulating images with natural language. In *NeurIPS*, 2018. 2

[14] Ethan Perez, Florian Strub, Harm de Vries, Vincent Dumoulin, and Aaron Courville. FiLM: Visual reasoning with a general conditioning layer. In *AAAI*, 2018. 2

[15] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. Learning transferable visual models from natural language supervision. In *ICML*, 2021. 2

[16] Aditya Ramesh, Prafull Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. Hierarchical text-conditional image generation with CLIP latents. In *arXiv:2204.06125*, 2022. 2

[17] Jheng-Wei Su, Hung-Kuo Chu, and Jia-Bin Huang. Instance-aware image colorization. In *CVPR*, 2020. 1, 2, 3, 6

[18] Tomihisa Welsh, Michael Ashikhmin, and Klaus Mueller. Transferring color to greyscale images. In *ACM TOG (SIGGRAPH)*, 2002. 1, 2

[19] Richard Zhang, Phillip Isola, and Alexei A. Efros. Colorful image colorization. In *ECCV*, 2016. 1, 2, 3